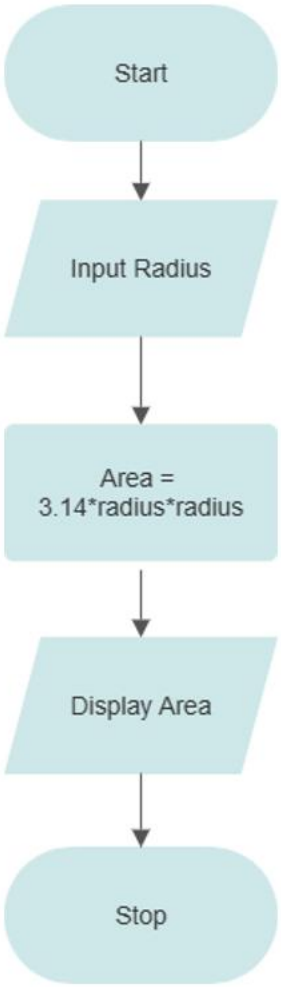
Experiment 1.1

Algorithm to Find Area of a Circle

Step 1: Start
Step 2: Read radius r
Step 3: Area = 3.14 × r. × r
Step 4: Display area
Step 5: Stop

Flowchart:



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1.1.1. Area of Circle

14:02

Write a Python program that calculates the area of a circle when the radius is provided by the user. Use π = 3.14 and display the area.

Input Format:

- A single line containing a floating-point number representing the radius.

Output Format:

- Print the computed area of the circle formatted to 4 decimal places.

Sample Test Cases +

Explorer

circlearea...

Submit

```
radius = float(input())
area = 3.14*radius*radius
print(f"{area:.4f}")
```

Debugger

Average time	Maximum time
0.010 s	0.013 s
9.75 ms	13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 13 ms

Debug

Expected output	Actual output
3.36	3.36
35.4493	35.4493

Test case 2 11 ms

Terminal Test cases

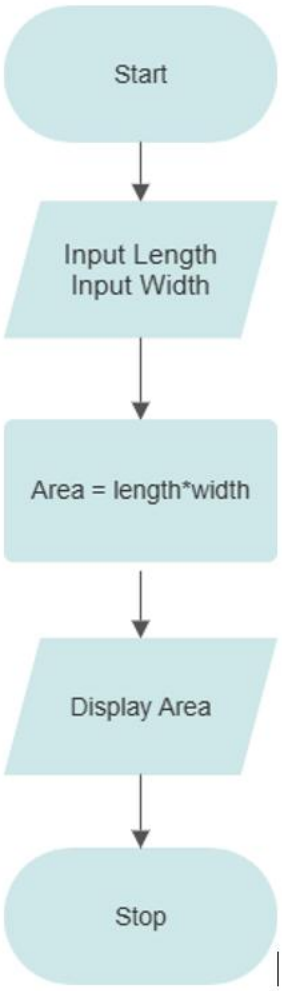
‹ Prev Reset Submit Next ›

Experiment 1.2 :

Algorithm to Find Area of a Rectangle

Step 1: Start
Step 2: Read length and width
Step 3: Area = length × width
Step 4: Display area
Step 5: Stop

Flowchart:



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1.1.2. Area of Rectangle

04:16

Write a Python program to calculate the area of a rectangle given its length and width.

Formula:

$$\text{Area of Rectangle} = \text{Length} \times \text{Width}$$

Input Format:

- First line contains a float value representing the length of the rectangle
- Second line contains a float value representing the width of the rectangle

Output Format:

- Print the area of the rectangle as a float value formatted to 2 decimal places.

Sample Test Cases +

Explorer

areaOfRe...

Submit

```python
# Type Content here...
length = float(input())
width = float(input())
area = length * width
print(f"{area:.2f}")
```

Average time	Maximum time
0.012 s	0.016 s
12.30 ms	16.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 16 ms

Debug

Expected output	Actual output
10.5	10.5
5.2	5.2
54.60	54.60

Test case 2 15 ms

Terminal Test cases

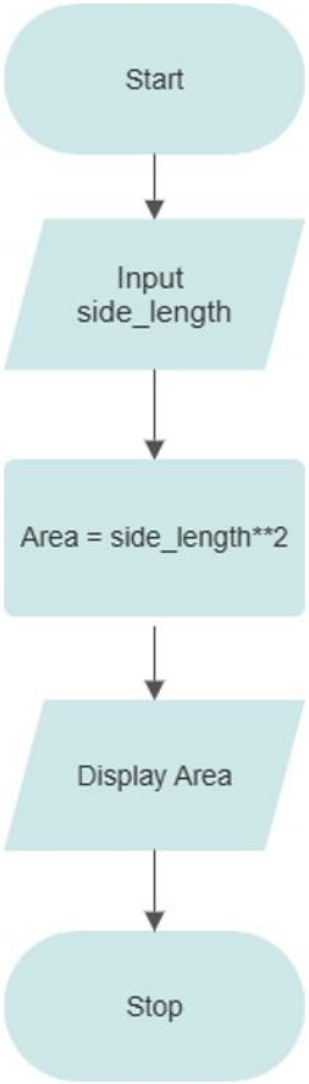
Debugger

Experiment 1.3

Algorithm to Find Area of a Square

Step 1: Start
Step 2: Read side_length
Step 3: Area = $(\text{side_length})^2$
Step 4: Display area
Step 5: Stop

Flowchart:



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1.1.3. Calculate Area of the Square

04:14

Write a Python program that prompts the user to enter the *side_length* of a square and computes the area of the square.

Formula:

- $\text{Area} = \text{side_length}^2$

Input Format:

- The input is a positive integer value that represents the *side_length* of the square.

Output Format:

- The output is a positive integer value that represents the area of the square.

Sample Test Cases +

Explorer

AreaSqua...

Submit

```
# Write your code here...
side_length = int(input())
area = side_length * side_length
print(area)
```

Debugger

Average time	Maximum time
0.009 s	0.013 s
9.00 ms	13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 13 ms

Debug

Expected output	Actual output
5	5
25	25

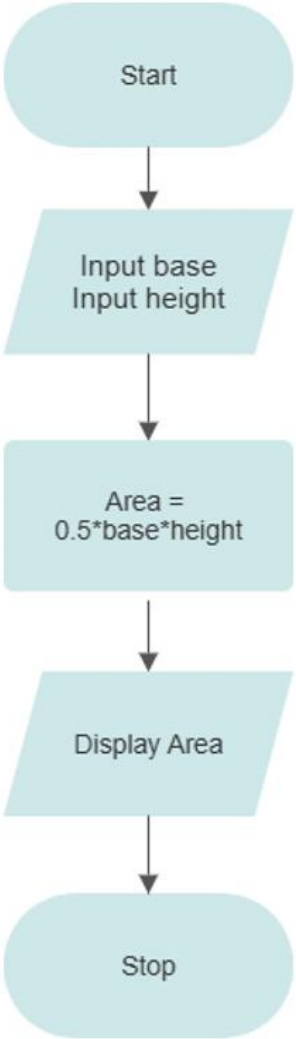
Test case 2 6 ms

Experiment 1.4

Algorithm to Find Area of a Triangle

Step 1: Start
Step 2: Read base and height
Step 3: Area = ½ × base × height
Step 4: Display area
Step 5: Stop

Flowchart:



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1.1.4. Area of Triangle

16:11

Write a Python program that prompts the user to enter the triangle's base and height and computes the triangle's area.

Formula: $Area\ of\ Triangle\ =\ 0.5\ \times base\ \times height.$

Input Format:

- The first line of input is the float value that represents the base of the triangle.
- The second line of input is the float value that represents the height of the triangle.

Output Format:

- The output is the floating point value that represents the area of a triangle, formatted to two decimals.

Sample Test Cases +

Explorer

triangleA...

Submit

```
base = float(input())
height = float(input())

area = 0.5 * base * height
print(f"{area:.2f}")
```

Debugger

Average time	Maximum time
0.015 s	0.023 s
15.00 ms	23.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 23 ms

Debug

Expected output	Actual output
6.54	6.54
1.23	1.23
4.02	4.02

Test case 2 8 ms

Terminal Test cases

‹ Prev Reset Submit Next ›

Experiment 1.5

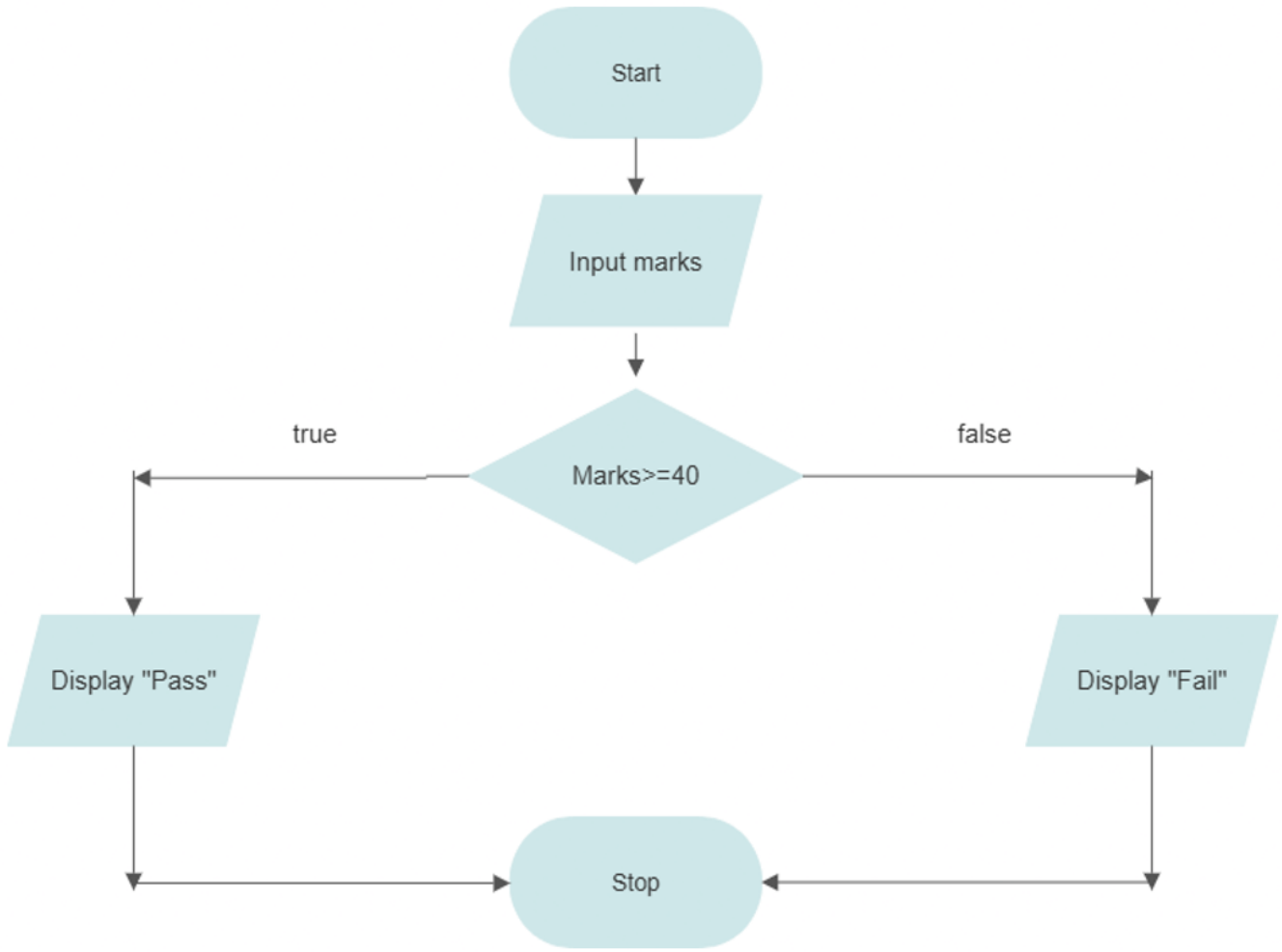
Algorithm to Determine Whether a Student Passed the Exam or Not

Step 1: Start
Step 2: Read marks
Step 3:

- If marks ≥ 40, display PASS
- Else, display FAIL

Step 4: Stop

Flowchart:



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1.1.5. Student Pass or Fail Status

08:09

Write a Python program to determine whether a student passed the exam or not based on their marks.

Pass/Fail Criteria:

- A student passes if marks ≥ 40
- A student fails if marks < 40

Input Format:

- Single line contains an integer representing the marks obtained by the student.

Output Format:

- Print "Pass" if the student passed the exam.
- Print "Fail" if the student failed the exam.

Sample Test Cases +

Explorer

passOrFa...

```python
marks = int(input())

if marks >= 40:
    print ("Pass")

else:
    print ("Fail")
```

Average time	Maximum time
0.007 s	0.010 s
6.71 ms	10.00 ms

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 7 ms

Debug

Expected output	Actual output
45	45
Pass	Pass

Test case 2 5 ms

Test case 3 8 ms

Terminal Test cases